

Liyan Xie

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RESEARCH AREA

- **Statistical Machine Learning:** Diffusion generative models, large language models, transfer learning, differential privacy, adversarial robustness.
- **Statistical Inference:** Time series analysis, survival analysis, sequential testing and detection.
- **Applications:** Healthcare, spatio-temporal data analysis, community detection, quality control.

EDUCATION

Ph.D.	Industrial Engineering <i>Major in Statistics</i>	School of Industrial and Systems Engineering Georgia Institute of Technology	2016 - 2021
B.S.	Statistics	University of Science and Technology of China	2012 - 2016

PROFESSIONAL APPOINTMENTS

2024.08 - present	Assistant Professor	Department of Industrial and Systems Engineering University of Minnesota
2021.09 - 2024.07	Assistant Professor	School of Data Science The Chinese University of Hong Kong, Shenzhen
2023.09 - 2024.02	Visiting Fellow <i>Hosted by Vincent Poor</i>	Department of Electrical and Computer Engineering Princeton University
2022.09 - 2022.12	Visiting Scholar <i>Hosted by Xiuyuan Cheng</i>	Department of Mathematics Duke University

PUBLICATION LIST

Journal Publications

- [J1] Yidong Ouyang, Liyan Xie, Guang Cheng, and Hongyuan Zha. “Alignment of Diffusion Model and Flow Matching for Text-to-Image Generation.” Accepted to *Transactions on Machine Learning Research (TMLR)*, 2026.
- [J2] Lixing Zhang, Xuran Liu, Ruizhi Zhang, and Liyan Xie. “Sequential Change Detection with Local Differential Privacy.” *Entropy, Special Issue on Information-Theoretic Security and Privacy*, 2026.
- [J3] Liyan Xie and Ruizhi Zhang. “Sequential Change Detection with Differential Privacy.” *IEEE Transactions on Information Theory*, 2025.
- [J4] Mohamed Seif, Liyan Xie, Andrea J. Goldsmith, H. Vincent Poor. “Private Online Community Detection for Censored Block Models.” *IEEE Transactions on Information Forensics & Security*, 2025.
- [J5] Liyan Xie, Xi He, Pinar Keskinocak, and Yao Xie. “Survival Analysis with Graph-Based Regularization for Predictors.” *Statistics in Biosciences*, 2025.
- [J6] Jingge Wang, Liyan Xie, Yao Xie, Shao-Lun Huang, Yang Li. “Generalizing to unseen domains with Wasserstein distributional robustness under limited source knowledge.” *IEEE Journal of Selected Topics in Signal Processing*, 2024.

- [J7] Xiaojun Zheng, Simon Mak, Liyan Xie, and Yao Xie. “eRPCA: Robust Principal Component Analysis for Exponential Family Distributions.” *Statistical Analysis and Data Mining: The ASA Data Science Journal*, 2024.
- [J8] Liyan Xie, Yao Xie, and George Moustakides. “Window-limited CUSUM for Sequential Change Detection.” *IEEE Transactions on Information Theory*, 2023.
- [J9] Minghe Zhang, Liyan Xie, and Yao Xie. “Spectral CUSUM for Online Network Structure Change Detection.” *IEEE Transactions on Information Theory*, 2023.
- [J10] Xiaojun Zheng, Simon Mak, Liyan Xie, and Yao Xie. “Percept: A New Online Change-Point Detection Method using Topological Data Analysis.” *Technometrics*, 2022.
- [J11] Haoyun Wang, Liyan Xie, Alex Cuzzo, Simon Mak, and Yao Xie. “Sequential Change-Point Detection for Mutually Exciting Point Processes.” *Technometrics*, 2022.
- [J12] Shixiang Zhu, Alexander Bukharin, Liyan Xie, Shihao Yang, Pinar Keskinocak, and Yao Xie. “Early Detection of COVID-19 Hotspots Using Spatio-Temporal Data.” *IEEE Journal of Selected Topics in Signal Processing*, 2022.
- [J13] Liyan Xie, Shaofeng Zou, Yao Xie, and Venugopal V. Veeravalli. “Sequential Change Detection: Classical Results and New Directions.” *IEEE Journal on Selected Areas in Information Theory*, 2021.
- [J14] Liyan Xie and Yao Xie. “Sequential Change Detection by Optimal Weighted ℓ_2 Divergence.” *IEEE Journal on Selected Areas in Information Theory*, 2021.
- [J15] Shixiang Zhu, Alexander Bukharin, Liyan Xie, Mauricio Santillana, Shihao Yang, and Yao Xie. “High-Resolution Spatio-Temporal Model for County-level COVID-19 Activity in the US.” *ACM Transactions on Management Information Systems*, 2021.
- [J16] Rios, Renee T., Francesca Lolli, Liyan Xie, Yao Xie, and Kimberly E. Kurtis. “Screening Candidate Supplementary Cementitious Materials under Standard and Accelerated Curing Through Time-series Surface Resistivity Measurements and Change-Point Detection.” *Cement and Concrete Research*, 2021.
- [J17] Anatoli Juditsky, Arkadi Nemirovski, Liyan Xie, and Yao Xie. “Convex Parameter Recovery for Interacting Marked Processes.” *IEEE Journal on Selected Areas in Information Theory*, vol. 1, no. 3, pp. 799-813, 2020.
- [J18] Liyan Xie, George V. Moustakides, and Yao Xie. “Sequential Subspace Changepoint Detection.” *Sequential Analysis*, vol. 39, no. 3, pp. 307-335, 2020. (Finalist of INFORMS QSR Best Student Paper Award 2019.)
- [J19] Yang Cao, Liyan Xie, Yao Xie, and Huan Xu. “Sequential Change-Point Detection via Online Convex Optimization.” *Entropy*, vol. 20, no. 2, pp. 108, 2018.

Refereed Conference and Workshop Papers

- [C1] Zewei Deng, Muhammad Siddeek, Liyan Xie, Mohamed Seif, Mengdi Wang, H. Vincent Poor, and Andrea Goldsmith. “Detecting Post-generation Edits to Watermarked LLM Outputs via Error-Correcting Codes.” *Findings of EMNLP*, 2026. (Short Paper)
- [C2] Lixing Zhang, Yidong Ouyang, Weifu Li, Shixiang Zhu, Guang Cheng, and Liyan Xie. “Learning What Not to Impute: An Uncertainty-Aware Diffusion Framework for Meaningful Missingness.” *Conference on Neural Information Processing Systems (NeurIPS)*, 2026.
- [C3] Liyan Xie and Martina Cardone. “Quickest Change Detection in Parametric Models With 1-Bit Measurements.” Accepted to *IEEE Information Theory Workshop (ITW)*, 2026.

- [C4] Zhengyan Wan, Yidong Ouyang, Liyan Xie, Fang Fang, Hongyuan Zha, and Guang Cheng. “Corrected Samplers for Discrete Flow Models.” *International Conference on Machine Learning (ICML)*, 2026.
- [C5] Zhengyan Wan, Yidong Ouyang, Qiang Yao, Liyan Xie, Fang Fang, Hongyuan Zha, and Guang Cheng. “Error Analysis of Discrete Flow with Generator Matching.” *International Conference on Machine Learning (ICML)*, 2026.
- [C6] Lixing Zhang, Liyan Xie, and Ruizhi Zhang. “Sequential Change Detection for Multiple Data Streams with Differential Privacy.” *IEEE International Symposium on Information Theory (ISIT)*, 2026.
- [C7] Zhengyan Wan, Yidong Ouyang, Liyan Xie, Fang Fang, Hongyuan Zha, and Guang Cheng. “Discrete Guidance Matching: Exact Guidance for Discrete Flow Matching.” *International Conference on Learning Representations (ICLR)*, 2026.
- [C8] Wenbin Zhou, Liyan Xie, Zhigang Peng, and Shixiang Zhu. “Sequential Change Point Detection via Denoising Score Matching.” *Allerton Conference on Communication, Control, and Computing*, 2025.
- [C9] Aoran Zhang, Wenbin Zhou, Liyan Xie, Shixiang Zhu. “Recurrent Neural Goodness-of-Fit Test for Time Series.” *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2025.
- [C10] Yidong Ouyang, Liyan Xie, Hongyuan Zha, Guang Cheng. “Transfer Learning for Diffusion Models.” *Conference on Neural Information Processing Systems (NeurIPS)*, 2024.
- [C11] Liyan Xie, Yuchen Liang, Venugopal V. Veeravalli. “Distributionally Robust Quickest Change Detection using Wasserstein Uncertainty Sets.” *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024.
- [C12] Yiran Yang and Liyan Xie. “Sequential Wasserstein Uncertainty Sets for Minimax Robust Online Change Detection.” *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*, 2024.
- [C13] Yidong Ouyang, Liyan Xie, and Guang Cheng. “Improving Adversarial Robustness by Contrastive Guided Diffusion Process.” *International Conference on Machine Learning (ICML)*, 2023.
- [C14] Yidong Ouyang, Liyan Xie, Chongxuan Li, and Guang Cheng. “MissDiff: Training Diffusion Models on Tabular Data with Missing Values.” *ICML Workshop on Structured Probabilistic Inference and Generative Modeling*, 2023.
- [C15] Yixuan Tan, Liyan Xie, and Xiuyuan Cheng. “Neural Differential Recurrent Neural Network with Adaptive Time Steps.” *NeurIPS Workshop on the Symbiosis of Deep Learning and Differential Equations (DLDE III)*, 2023.
- [C16] Shixiang Zhu, Liyan Xie, Minghe Zhang, Rui Gao, and Yao Xie. “Distributionally Robust weighted k-Nearest Neighbors.” *Conference on Neural Information Processing Systems (NeurIPS)*, 2022.
- [C17] Liyan Xie. “Minimax Robust Quickest Change Detection using Wasserstein Ambiguity Sets.” *IEEE International Symposium on Information Theory (ISIT)*, 2022.
- [C18] Liyan Xie and Yao Xie. “Optimality of Graph Scanning Statistic for Online Community Detection.” *IEEE International Symposium on Information Theory (ISIT)*, 2021.
- [C19] Haoyun Wang, Liyan Xie, Alex Cuzzo, Simon Mak, and Yao Xie. “Uncertainty Quantification for Inferring Hawkes Networks.” *Conference on Neural Information Processing Systems (NeurIPS)*, 2020.

- [C20] Minghe Zhang, Liyan Xie, and Yao Xie. “Online Community Detection by Spectral CUSUM.” *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2020.
- [C21] Liyan Xie, Yao Xie, and George V. Moustakides. “Asynchronous Multi-Sensor Change-Point Detection for Seismic Tremors.” *IEEE International Symposium on Information Theory (ISIT)*, pp. 787–791, 2019.
- [C22] Rui Gao, Liyan Xie, Yao Xie, and Huan Xu. “Robust Hypothesis Testing Using Wasserstein Uncertainty Sets.” *Advances in Neural Information Processing Systems (NeurIPS)*, pp. 7902–7912, 2018. (spotlight)
- [C23] Liyan Xie, George V. Moustakides, and Yao Xie. “First-Order Optimal Sequential Subspace Change-point Detection.” *IEEE Global Conference on Signal and Information Processing (GlobalSIP)*, pp. 111–115, 2018.
- [C24] Yang Cao, Liyan Xie, Yao Xie, and Huan Xu. “Nearly Second-Order Optimality of Online Joint Detection and Estimation via One-Sample Update Schemes.” *Proceedings of the Twenty-First International Conference on Artificial Intelligence and Statistics (AISTATS)*, vol. 84, pp. 519–528, 2018.
- [C25] Liyan Xie, Yao Xie, Sin-Mei Wu, Fan-Chi Lin, and WenZhan Song. “Communication Efficient Signal Detection for Distributed Ambient Noise Imaging.” *IEEE Asilomar Conference on Signals, Systems, and Computers*, pp. 1779–1783, 2018.
- [C26] Liyan Xie and Yao Xie. “Sequential Detection of Low-Rank Changes Using Extreme Eigenvalues.” *IEEE 7th International Workshop on Computational Advances in Multi-Sensor Adaptive Processing (CAMSAP)*, pp. 1–5, 2017.

HONORS AND AWARDS

2021	Robert Goodell Brown Fellowship for excellence in research (Data Science and Statistics), ISyE
2020	IDEaS-TRIAD and ARC-TRIAD Research Fellowships, Georgia Tech
2020	Selected to participate in EECS Rising Stars Workshop
2020	Outstanding Graduate Teaching Assistant Award, ISyE, Georgia Tech
2019	Runner up for INFORMS Computing Society Student Paper Prize
2019	Finalist of INFORMS QSR Best Student Paper Award
2019	Poster Award, Georgia Statistics Day
2019	Poster Award, WuFest Conference
2018	Poster Award, Georgia Statistics Day

PROFESSIONAL SERVICE

Associate Editor, Statistics and Computing, since 2024.

Journal Referee of IEEE Transactions on Signal Processing, Sequential Analysis, Electronic Journal of Statistics, Journal of the Royal Statistical Society: Series B, Journal of the American Statistical Association, IEEE Transactions on Information Theory, Management Science, etc.

Conference Reviewer for: ICML, NeurIPS, ICLR, AISTATS, AAAI, ISIT, etc.